A03 / INDIVIDUAL LAB REPORT

Neural Architecture & Noise Engineering

Neural Network Zoo • Neural Network Safari Reflection

Team Member Olu

Role Neural Architecture & Noise Engineering Lead

1. Introduction & Objectives

My focus for the Neural Network Safari assignment was constructing the core neural architecture blocks and defining mathematical noise schedules. Within our safari metaphor, this represents the internal mechanics and *metabolic engine* of our models.

My work centered on designing backbone networks, including U-Net components, convolutional layers, and skip connections, capable of transforming complex data distributions.

> **Primary objective**
> Develop the architectural building blocks and noise scheduling mechanisms that support the model's ability to learn complex data distributions.

2. Technical Execution & Methodology

My contributions targeted the structural and mathematical foundations of our models.

01 Backbone Construction

Coded fundamental U-Net encoder-decoder blocks featuring residual connections and time-embedding layers to preserve spatial hierarchies.

02 Skip-Connection Mapping

Implemented skip connections to bridge high-resolution encoder features with upsampling decoder layers, preventing spatial information loss.

03 Noise Scheduling β_t

Formulated linear and cosine variance schedules to govern the corruption process across discrete timesteps t.

Noise scheduling principle
The variance schedule controls the amount of noise introduced at each timestep. In a standard diffusion formulation, the forward process uses

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\right),$$

where β_t determines the variance of the noise added at timestep t.

04 Loss Function Implementation

Integrated mean squared error (MSE) loss formulations to evaluate the network's ability to predict injected noise at arbitrary timesteps.

The noise-prediction objective can be expressed as

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right],$$

where ϵ is the injected noise and ϵ_θ is the network's noise prediction.

3. Challenges & Solutions

Challenge: Gradient-flow stability
Ensuring gradient flow stability through deep network layers without encountering vanishing or exploding gradients.

Solution: Architectural stabilization and testing
Integrated layer normalization and residual scaling within the custom architectural blocks, verifying gradient health through unit tests.

4. Reflection

Designing these architectures deepened my understanding of how structural inductive biases, such as translation invariance in convolutions and skip connections in U-Nets, dictate a network's capacity to learn specific data distributions.

Working closely with Cherluk's data loaders and Taki's sampling loops ensured our model backbone performed efficiently under experimental conditions.

Key insight
Architecture determines how information is represented; noise engineering determines the corruption process the model must learn to reverse.
Reliable model behavior depends on aligning these components with the data pipeline and sampling procedure. The lab reinforced the importance of treating architecture, optimization, and diffusion scheduling as interconnected parts of one system.



OLU | A03 NEURAL NETWORK ZOO

Neural Architecture • Noise Engineering • Collaborative Model Development